

Intel Image Classification

A CNN built from scratch to classify natural scene images into 6 categories using the Intel Image Classification dataset.

~25,000 training images

6 scene categories

150 × 150 px input

3 conv blocks

Kaggle dataset

DATASET — 6 SCENE CLASSES

Buildings

Urban structures

Forest

Tree canopies

Glacier

Ice & snow

Mountain

Rock terrain

Sea

Ocean & water

Street

Road scenes

END-TO-END PIPELINE

01

Import Libraries

TF/Keras, sklearn, matplotlib, seaborn

02

Download Dataset

Kaggle API, ~350 MB zip

03

Explore Data

Class distribution, sample images

04

Preprocess & Augment

Rescale + 5 augmentations, 80/20 split

05

Build CNN

3× Conv + Dense + Dropout(0.5)

06

Compile

Adam · CatCrossEntropy · Accuracy

07

Train

20 epochs, EarlyStopping, Checkpoint

08-12

Evaluate

Curves · Report · Confusion · Preds

MODEL ARCHITECTURE

		Input block
Input	Image tensor	150×150×3
		Convolutional blocks ×3
Conv2D	32 filters, 3×3, ReLU	148×148×32
MaxPool	Pool 2×2	74×74×32
Conv2D	64 filters, 3×3, ReLU	72×72×64
MaxPool	Pool 2×2	36×36×64
Conv2D	128 filters, 3×3, ReLU	34×34×128
MaxPool	Pool 2×2	17×17×128
		Classifier head
Flatten	—	36,992
Dense	256 units, ReLU	256
Dropout	rate 0.5	—
Dense	6 units, Softmax	6

DATA AUGMENTATION (TRAINING ONLY)

Rescale

+255

Rotation range

20°

Zoom range

0.2×

Width shift

0.2

Height shift

0.2

Horizontal flip

ON

Val split

20%

TRAINING CALLBACKS

EarlyStopping

Halts training when validation loss stops improving. Restores the best weights automatically.

monitor: val_loss

patience: 5

restore_best_weights

ModelCheckpoint

Saves model weights to disk only when validation accuracy improves — guarantees best model is kept.

monitor: val_accuracy

save_best_only

KEY HYPERPARAMETERS

Image size

150²

Batch size

32

Max epochs

20

Dropout

0.5

Optimizer

Adam

Patience

5

EVALUATION METRICS USED

Accuracy / Loss

model.evaluate()

Precision

per class

Recall

per class

F1-Score

harmonic mean

Confusion matrix

heatmap

Visual preds

10 samples

CONFUSION MATRIX NOTES

Diagonal

Correct predictions

Off-diagonal

Misclassifications

Common error

Glacier ↔ Mountain (similar textures)

Visualization

Seaborn heatmap, Blues cmap

Test set is **separate** from validation — drawn from `seg_test/seg_test` , not seen during training.